DocsAI Standard Operating Procedure (SOP)

Version 1.0

1. Purpose

Define the standard workflow for document ingestion, AI processing, semantic indexing, callback handling, and persistence in DocsAI.

2. Scope

Applies to the Go backend, Python AI service, MinIO, PostgreSQL, Redis, gRPC communication, and future search/chat modules.

3. Architecture Overview

DocsAI consists of a Go backend that manages APIs, storage, metadata and orchestration, and a Python AI service that processes documents by loading, chunking and embedding them before sending results back through a reverse gRPC callback.

4. End-to-End Workflow

1. Upload document.
2. Store file in MinIO.
3. Save document metadata.
4. Invoke AI service via ProcessDocument gRPC.
5. AI downloads file.
6. Detect loader and extract content.
7. Chunk document.
8. Generate Gemini embeddings.
9. Build ProcessingResult.
10. Callback SaveProcessingResult to backend.
11. Persist chunks and embeddings.
12. Mark document as completed.

5. Components

Backend (Go), AI Service (Python), PostgreSQL, Redis, MinIO, Gemini API, gRPC.

6. Supported Formats

PDF, DOCX, XLSX, CSV, PPTX, TXT and images (OCR-ready).

7. Logging & Monitoring

Use structured logs, monitor processing status, callback success, and failures.

8. Security

Protect credentials, validate uploads, secure gRPC communication, and avoid committing secrets.

9. Future Roadmap

Semantic search, pgvector, RAG chat, OCR, knowledge graph, workflows, collections, workspaces, APIs and analytics.